

CH316 Final Exam Study Guide

The Final exam is meant to test your understanding of the objectives of the experiments you performed in the lab, the reason for major step in the procedure including instrument setup and operation where applicable, as well as data interpretation including calculations. The exam will include multiple choice, fill in the blank, short answer and calculation questions.

Review of calculations and questions included in the quizzes and in the lab report submissions should help you prepare for the final exam. Additional information is provided here to help you better prepare for the Final. In order to do well on the final exam, you should be able to:

Answer concept questions related to experiments performed in the lab. Examples include:

1. Identify buffer and equivalence point regions of a titration curve
2. Identify point on the titration curve of a weak acid with a strong base that corresponds to the pK_a of a weak acid
3. Describe the difference between a pH electrode and a fluoride ion selective electrode.
4. Define LOD and how can it be calculated. What was the order of magnitude of the limit of detection in the molecular fluorescence experiment (e.g. mg/mL, ng/mL, μ g/mL)?
5. Apply the quantitative laws of absorption and emission spectroscopy, chromatography, and Potentiometry
6. Recognize which technique generally offers a lower detection limit, molecular absorption or emission (fluorescence)?
7. What is a matrix effect? What type of calibration is best to deal with a matrix effect?
8. Describe the standard addition calibration method. How can one decide if standard addition should be used instead of the regular calibration method?
9. Describe the method of calibration with internal standard. How can one decide if an internal standard should be used?
10. Describe if a more polar compound will elute earlier or later than a less polar compound on a C18 column. How does increasing the %organic in the mobile phase affect the retention time of a component on a C18 column?
11. How does changing the column temperature affect retention times in GC?

Describe or give the reason for major steps in the lab procedures. Examples include:

1. Recognize appropriate tools for measuring mass and volume in quantitative analysis (analytical balance, volumetric glassware) and their proper use (e.g. handling weighing flasks with tongues and not blowing the last drop from a volumetric pipet)
2. Differentiate between/give examples of sources of random error and sources of systematic error in lab measurements.
3. Describe how to calibrate a volumetric pipet. What type of error does calibration get rid of (random or systematic)?

4. During an acid-base titration, if the titrant used is NaOH and the buret is washed with DI water but not conditioned with NaOH, will the calculated $[H^+]$ concentration be higher or lower than the true value?
5. Explain why TISAB is added to all fluoride standards and unknowns during the analysis of fluoride using an ion selective electrode? If TISAB was not added and the unknown solution is too basic (high pH) how will that affect the observed potential reading or the calculated fluoride concentration (higher or lower than true value?) ..etc.
6. Explain why was the molybdate reagent added to the phosphorus solutions in the molecular absorption experiment.
7. Explain why was H_2SO_4 added to all standards and unknowns during the fluorescence experiment.
8. Recognize that in all calibration experiments, dilution schemes of unknowns are devised such that the response from the unknown falls closer to the middle of the calibration curve.
9. Why was a potassium solution added to all standards and unknowns in the atomic absorption and emission experiments? Be able to answer related questions such as:
 - a. How would the detected signals from sodium emission and calcium absorption be impacted if the potassium solution were not added?
 - b. Does the presence of the ionization suppressor (K^+) change the ion/neutral fraction? Increase/decrease?
10. Explain why uracil was added to the steroid sample in chromatographic separation by HPLC?
11. Why was butylbenzene added to all limonene standard and unknown solutions in the GC experiment?

Perform solution calculations similar to those performed in the lab and lab report. Examples include:

1. Perform dilution and stoichiometric calculations to determine concentrations of stock and working standards.
2. Determine concentration of unknowns from quantitative laws of absorption, emission spectroscopy, potentiometry and chromatography.
3. Devise a dilution scheme for the preparation of an unknown solution to target the middle of a calibration curve (e.g. given a calibration curve, devise a dilution scheme to prepare a dilute soy sauce solution given the label Na content).

Perform data analysis calculations similar to those submitted in lab reports. Examples include:

1. Calculate standard error from averaging repeat measurements or from linear regression (calibration curve) calculations.
2. Recognize the output from Linest function in Excel and be able to use it in calculations.
3. Propagate error from measured concentration of dilute unknown to original sample (e.g. error in $\%P_2O_5$ in a fertilizer sample, $\%w/v$ fluoride in mouthwash, quinine in tonic water).

4. Calculate the 95% confidence limit ($CL_{\text{value}} = t * s_{\text{value}}$) in a reported value - select the appropriate degrees of freedom to find t based on repeat measurement or linear regression.
5. Report results to the correct number of significant figures based on the confidence limit.
6. Perform a t test to determine if an observed value is significantly different than an expected value at the 95% confidence level.
7. Calculate unknown concentration and standard error from linear regression for normal calibration curves with or without internal standard (e.g. calculations in Na emission and GC labs) and for calibration curves with standard addition (e.g. calculations in Ca absorption lab). For calculations involving s_c and s_{x0} , know what values go in the equations.